Map of 5 data collection areas in Madagascar

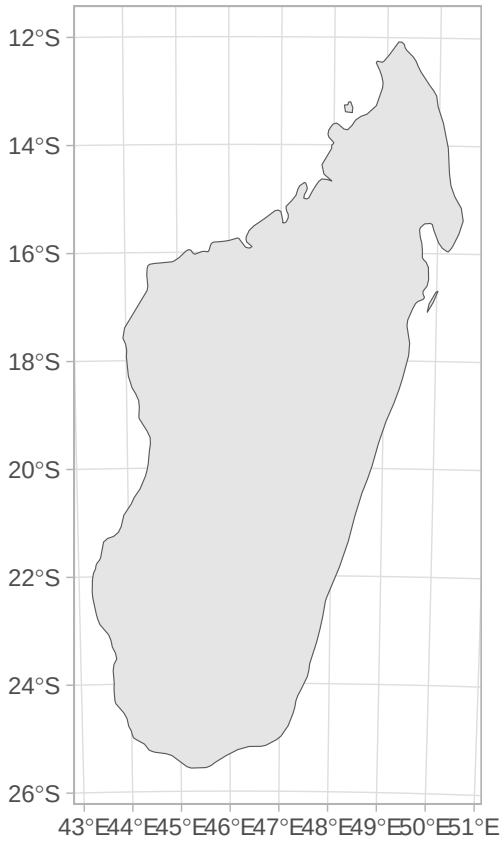
2024-02-07

```
res_dir <- "../results/s09_data_source_map/"
```

New plan for map with rnaturalearth

```
# WGS84 is standard, but lets use Laborde
Madagascar <- ne_countries(country = "Madagascar", type = "map_units", scale = "medium")

ggplot() +
  geom_sf(data = Madagascar) +
  coord_sf(crs = st_crs(29702)) + # Laborde projection, commonly used for Madagascar
  theme_light()
```

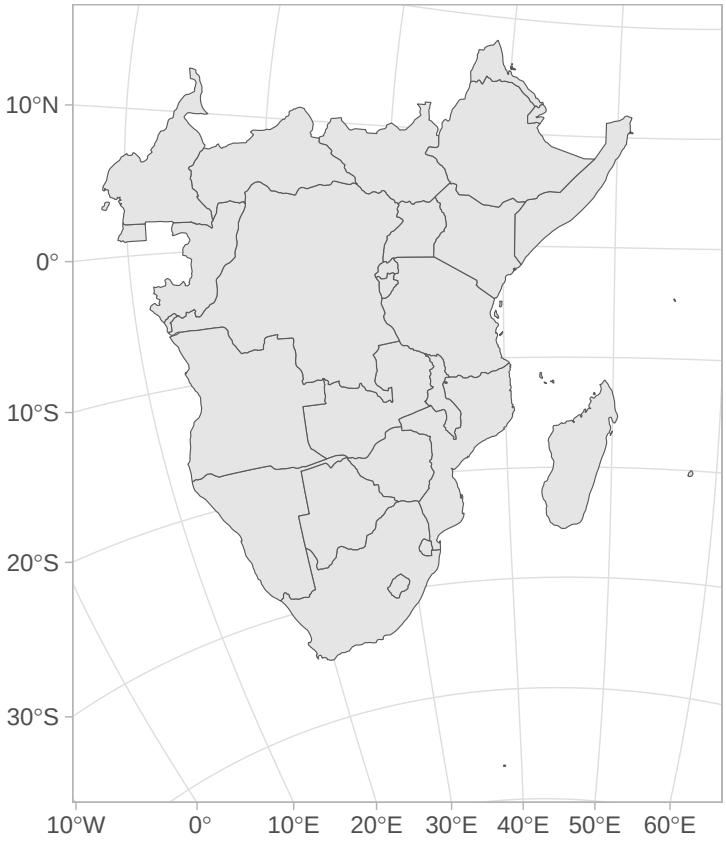


```
target <- c("Madagascar", "Mozambique", "Comoros", "Mauritius", "Seychelles", "Mayotte", "United Republi

# "Reunion" "Equatorial Guinea"!!


Madagascar_inset <- ne_countries(country = target, type = "map_units", scale = 50)

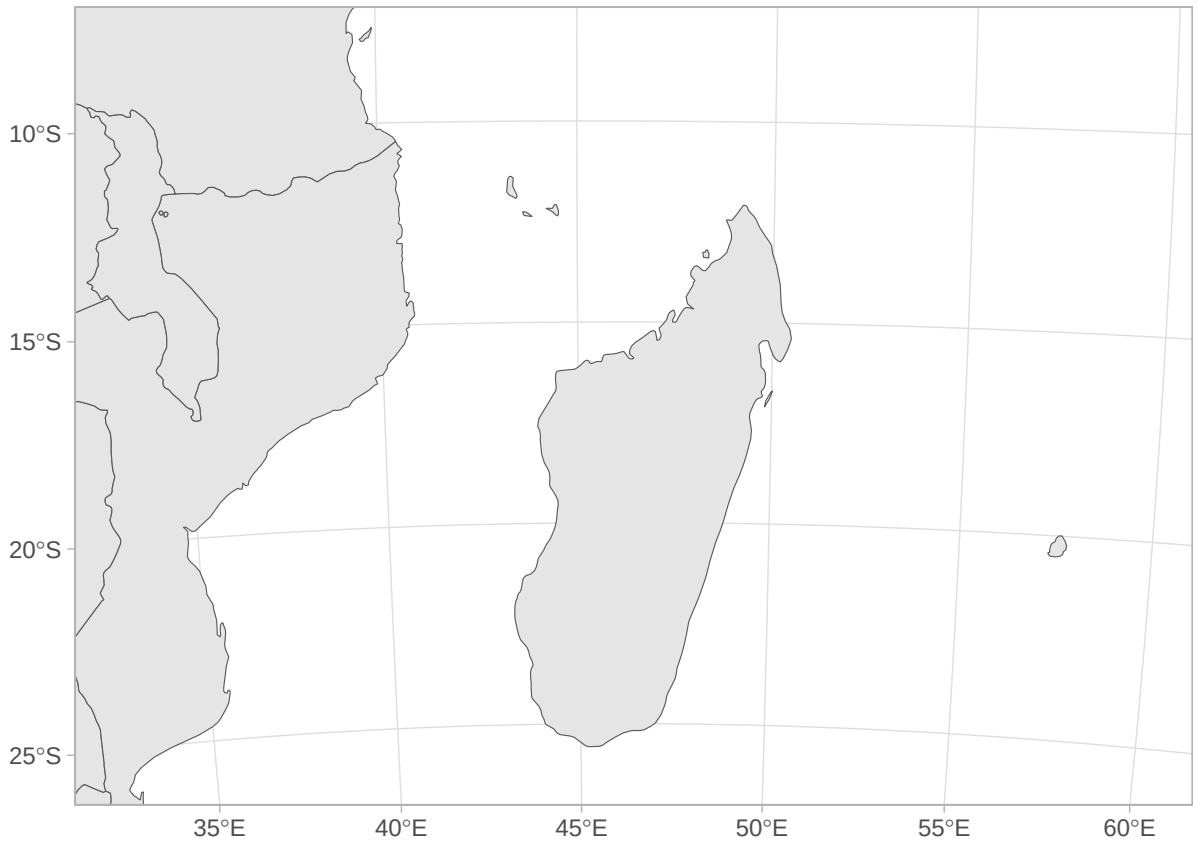
ggplot() +
  geom_sf(data = Madagascar_inset) +
  coord_sf(crs = st_crs(29702)) + # Laborde projection, commonly used for Madagascar
  theme_light()
```




```
# getting the right area is HARD lims_method = "cross" by default but will try others

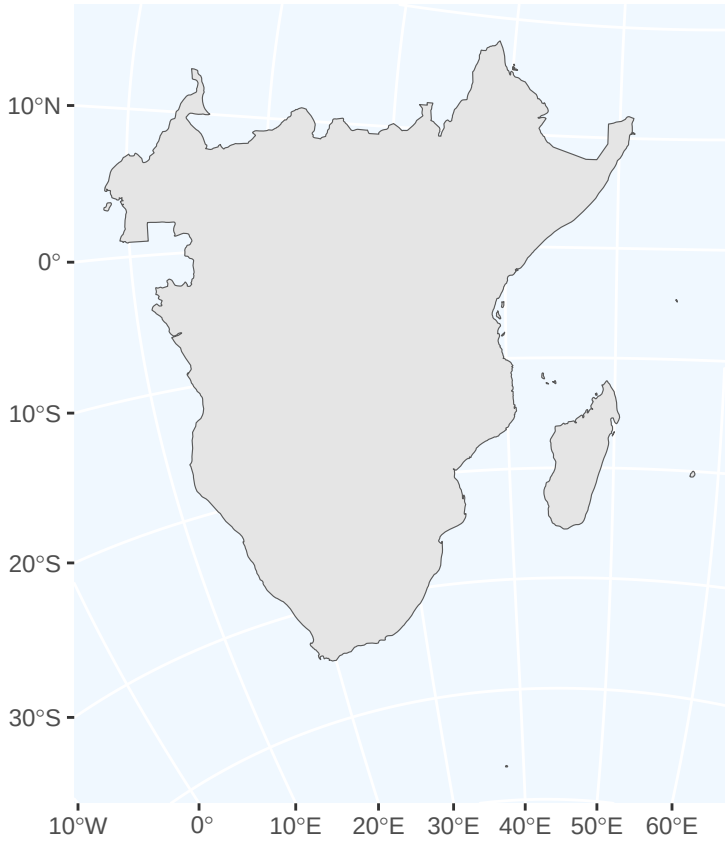
# others don't make much difference, confusingly
```

```
ggplot() +
  geom_sf(data = Madagascar_inset) +
  coord_sf(crs = st_crs(29702), xlim = c(-1000000, 1800000), ylim = c(1, 2000000)) + # Laborde projecti
  theme_light()
```



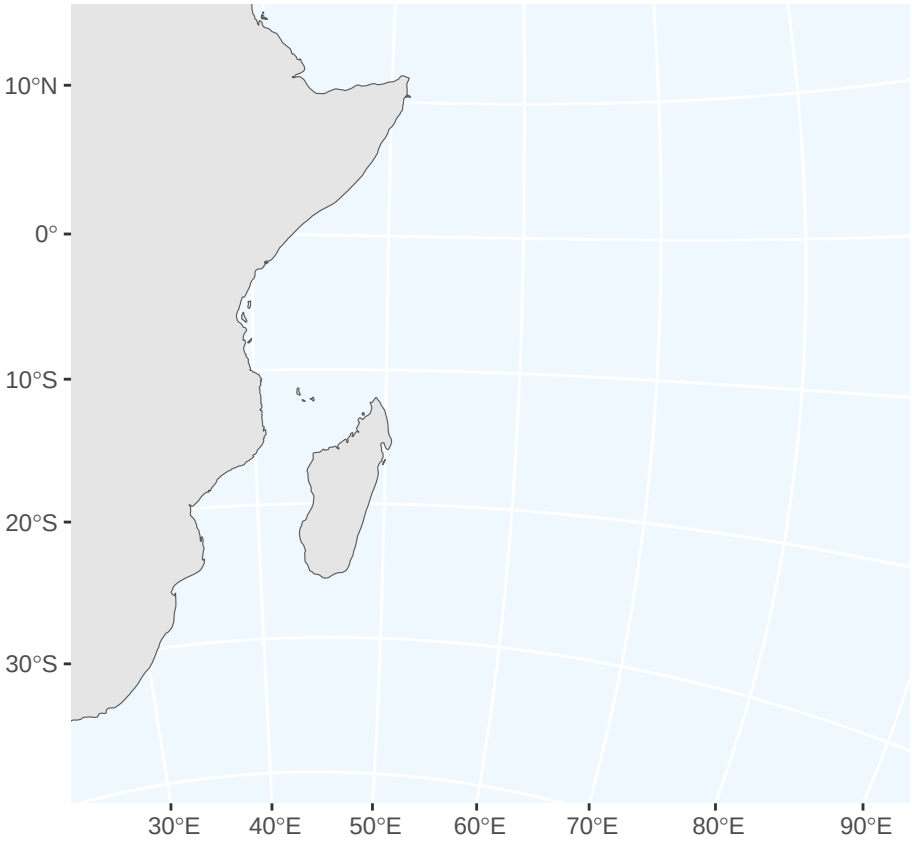

```
# Dissolve boundaries into a single geometry
Madagascar_dissolved <- Madagascar_inset %>%
  st_union() %>%
  st_as_sf() # Convert back to sf object

# Plot the map without boundaries
ggplot() +
  geom_sf(data = Madagascar_dissolved) +
  theme(panel.background = element_rect(fill = "#F0F8FF", color = NA)) +
  coord_sf(crs = st_crs(29702)) # Laborde projection
```




```r
# Get all African countries
africa <- ne_countries(continent = "Africa", scale = 50, returnclass = "sf") %>%
  st_union() %>%
  st_as_sf()

# Plot
ggplot() +
  geom_sf(data = africa) +
  coord_sf(crs = st_crs(29702), xlim = c(-1500000, 4800000), ylim = c(-1500000, 4500000)) +
  theme(panel.background = element_rect(fill = "#F0F8FF", color = NA))
```




```
# kml_data <- sf::st_read(dsn = "../data/raw/data_input/Data_with_pierron.kml")
# kml_data <- sf::st_transform(kml_data, crs = 29702)
#
# # Add names
# kml_data$Name <- c("Merina", "Sakalava", "Betsimisaraka", "Betsileo", "Diego", "Temoro", "Vezo", "Mik
#
# kml_data <- kml_data %>%
#   mutate(
#     x = st_coordinates(.)[, 1],
#     y = st_coordinates(.)[, 2]
#   )



kml_data <- readRDS("../data/raw/data_input/sampling_locations_with_pierron.rds")

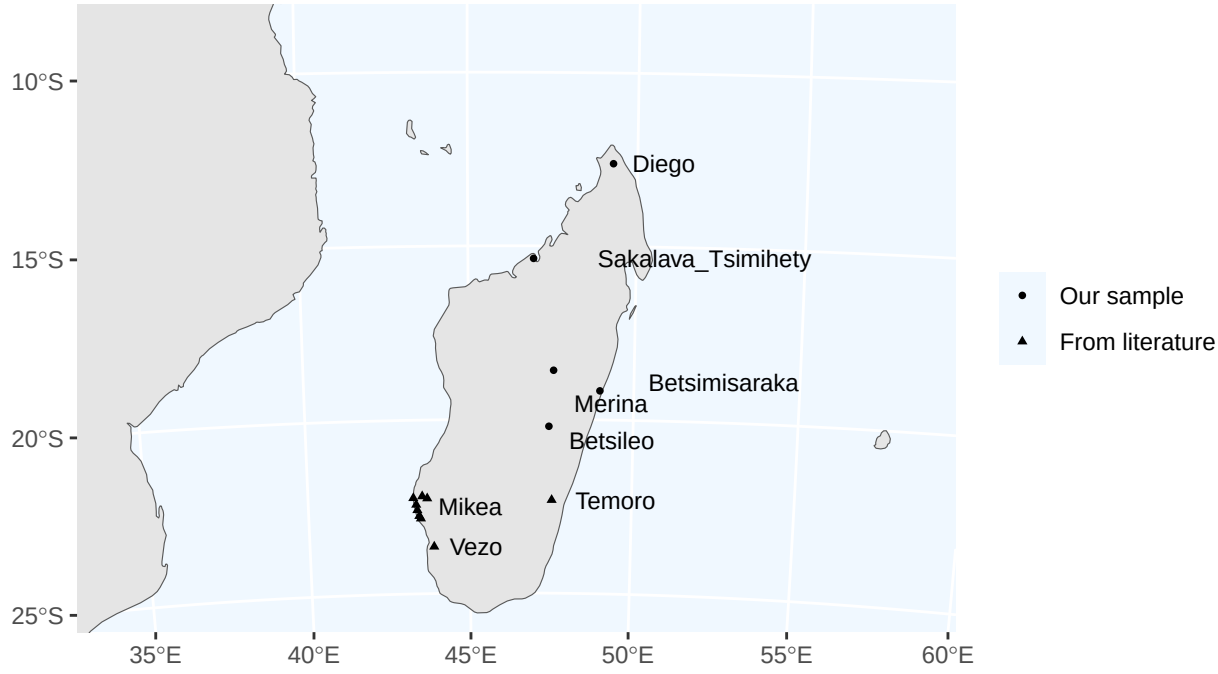

ours <- rep(0, 5)
pierron <- rep(1, 9)
sample <- c(ours, pierron)
kml_data$sample <- sample

labels_data <- kml_data %>%
  group_by(Name) %>%
  summarise(
    x = mean(x),
```

```
    y = mean(y),
    geometry = st_centroid(st_union(geometry)) # Optional: Keeps geometry if needed
  )

# Replace kml_data$Name == "Sakalava_Tsimihety" with "Sakalava/Tsimihety"
kml_data$Name[kml_data$Name == "Sakalava_Tsimihety"] <- "Sakalava/Tsimihety"
# Replace kmldata$Name == "Diego" with "Diego Mixed"
kml_data$Name[kml_data$Name == "Diego"] <- "Diego Mixed"

ggplot() +
  geom_sf(data = Madagascar_dissolved) +
  geom_sf(data = sf::st_as_sf(kml_data), aes(shape = factor(sample)), size = 1) +
  coord_sf(crs = st_crs(29702), xlim = c(-1000000, 1800000), ylim = c(1, 2000000), expand = FALSE) + # A
  theme(panel.background = element_rect(fill = "#F0F8FF", color = NA)) +
  scale_shape_manual(values = c(16, 17), labels = c("Our sample", "From literature")) +
  labs(shape = NULL) +
  # geom_text(data = kml_data, aes(x = x, y = y, label = Name), size = 3, hjust = -0.3)
  geom_text(data = labels_data, aes(x = x, y = y, label = Name), size = 3, hjust = -0.3) +
  # geom_text_repel(data = labels_data, aes(x = x, y = y, label = Name), max.overlaps = 10, force = 200
  labs(x = NULL, y = NULL)
```




```
ggsave(paste0(res_dir, "madagascar_map_labels.jpg"), width = 7, height = 7, dpi = 300)
```

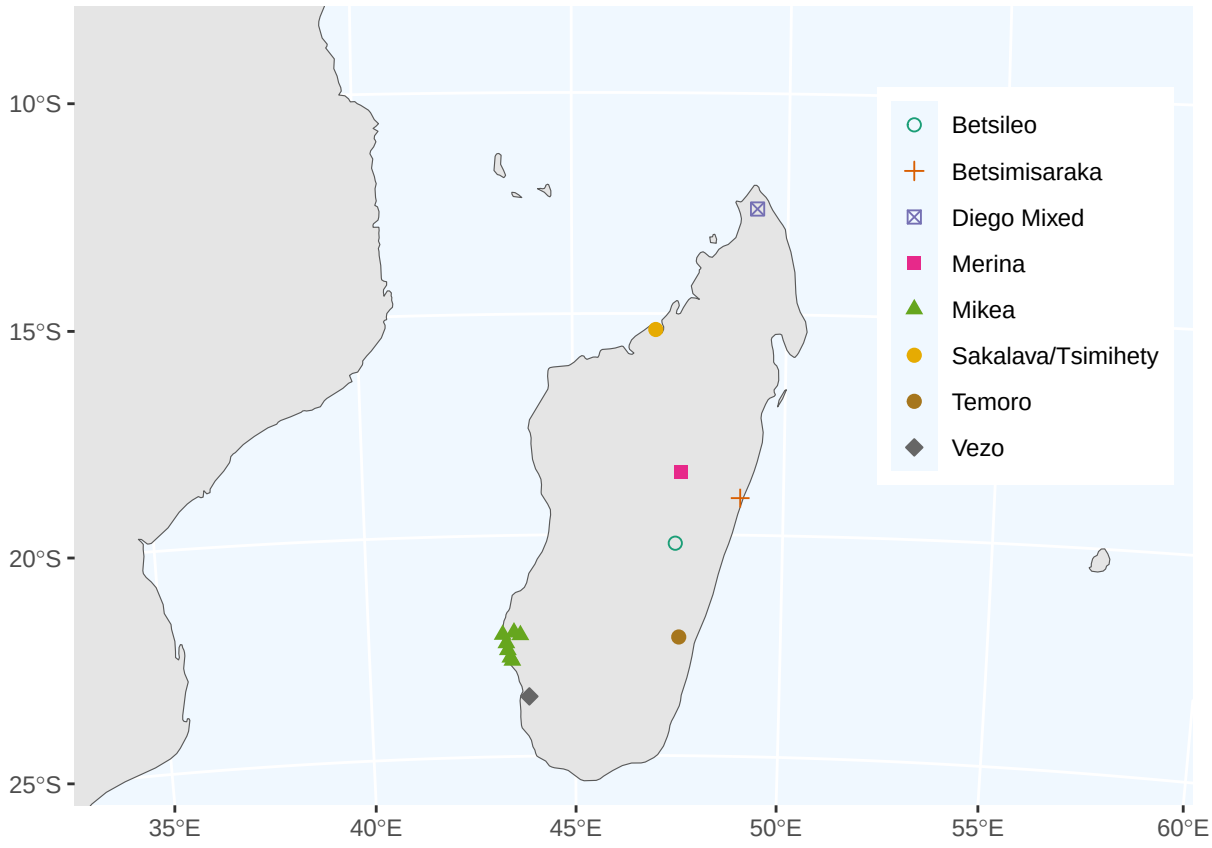
```r
kml_data <- kml_data[order(kml_data$Name), ]

kml_data$shape <- c(1, 3, 7, 15, 17, 17, 17, 17, 17, 17, 17, 19, 21, 23)

main_map <- ggplot() +
  geom_sf(data = Madagascar_dissolved) +
  geom_sf(data = sf::st_as_sf(kml_data), aes(shape = Name, color = Name, fill = Name), size = 2) +
  coord_sf(crs = st_crs(29702), xlim = c(-1000000, 1800000), ylim = c(1, 2000000), expand = FALSE) + # 
  theme(panel.background = element_rect(fill = "#F0F8FF", color = NA)) +
  scale_shape_manual(values = c(1, 3, 7, 15, 17, 19, 21, 23)) +
  labs(shape = NULL) +
  # geom_text(data = labels_data, aes(x = x, y = y, label = Name), size = 3, hjust = -0.3)+
  labs(x = NULL, y = NULL, color = NULL, shape = NULL, fill = NULL) +
  scale_color_brewer(palette = "Dark2") +
  scale_fill_brewer(palette = "Dark2") +
  guides() +
  theme(  legend.position = c(0.85, 0.65),
    #legend.position = "right",
    legend.background = element_rect(fill = "white")
  )
```

```
## Warning: A numeric ‘legend.position‘ argument in ‘theme()‘ was deprecated in ggplot2
## 3.5.0.
## i Please use the ‘legend.position.inside‘ argument of ‘theme()‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```

```r
main_map
```




```
ggsave(paste0(res_dir, "madagascar_map_pca_symbols.jpg"), width = 7, height = 5, dpi = 300)
```

Create an inset

```
library(ggplot2)
library(sf)
library(cowplot)
library(rnaturalearth)
library(rnaturalearthdata)
```

```
##
## Attaching package: 'rnaturalearthdata'

## The following object is masked from 'package:rnaturalearth':
##
##     countries110
```

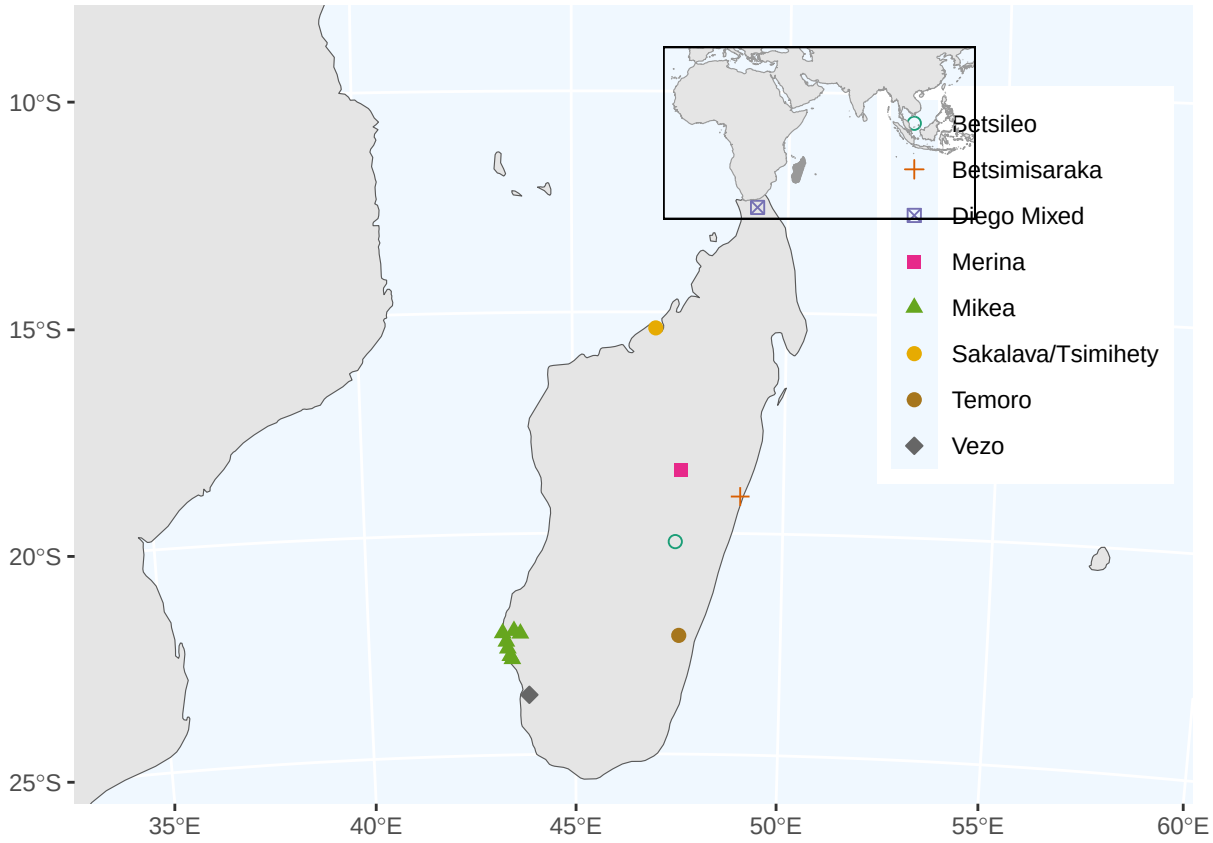
```
africa <- ne_countries(scale = "medium", continent = c("Africa", "Asia", "Europe"), returnclass = "sf")
madagascar_outline <- ne_countries(scale = "medium", country = "Madagascar", returnclass = "sf")

inset_map <- ggplot() +
  geom_sf(data = africa, fill = "gray90", color = "gray60") +
```

```
  geom_sf(data = madagascar_outline, fill = "gray60", color = "gray60") +
  coord_sf(xlim = c(-15, 130), ylim = c(-40, 40)) +
  theme_void() +
  theme(panel.border = element_rect(color = "black", fill = NA, linewidth = 0.5))

# --- Combine main map and inset ---
final_plot <- ggdraw() +
  draw_plot(main_map) +
  draw_plot(inset_map, x = 0.55, y = 0.71, width = 0.25, height = 0.25) # adjust position/size

final_plot
```

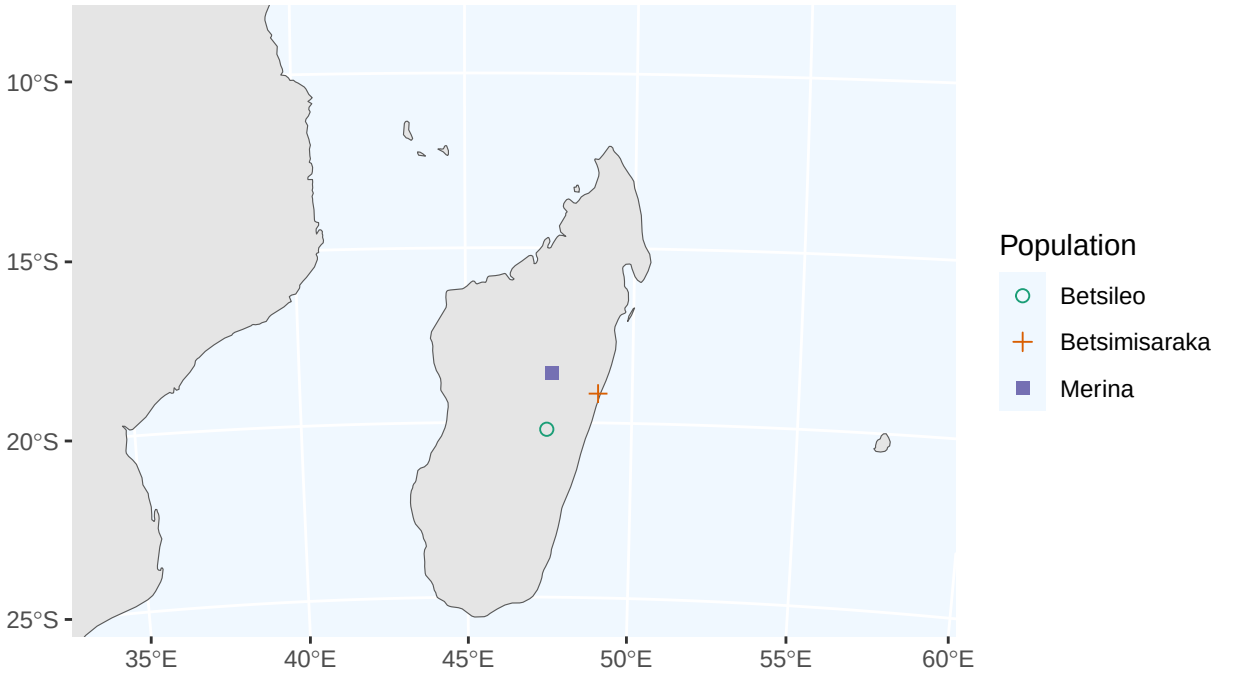



```
ggsave(paste0(res_dir, "madagascar_map_pca_symbols_inset.jpg"), width = 7, height = 5, dpi = 300)
```

Extra map for poster

```
# filter kml data to only betsileo, merina, betsimisaraka, and sakalava
kml_data_subset <- kml_data[kml_data$Name %in% c("Betsileo", "Merina", "Betsimisaraka", "Sakalava"), ]
# rename "sakalava" to "Sakalava/Tsimihety
kml_data_subset$Name[kml_data_subset$Name == "Sakalava"] <- "Sakalava/Tsimihety"
ggplot() +
  geom_sf(data = Madagascar_dissolved) +
```

```
  geom_sf(data = sf::st_as_sf(kml_data_subset), aes(shape = Name, color = Name, fill = Name), size = 2)
  coord_sf(crs = st_crs(29702), xlim = c(-1000000, 1800000), ylim = c(1, 2000000), expand = FALSE) + # 
  theme(panel.background = element_rect(fill = "#F0F8FF", color = NA)) +
  scale_shape_manual(values = c(1, 3, 15, 19)) +
  labs(shape = NULL) +
  # geom_text(data = labels_data, aes(x = x, y = y, label = Name), size = 3, hjust = -0.3)+
  labs(x = NULL, y = NULL, color = "Population", shape = "Population", fill = "Population") +
  # scale_color_brewer(palette = "Dark2")+
  scale_color_manual(values = c("Betsileo" = "#1B9E77", "Merina" = "#7570B3", "Betsimisaraka" = "#D95F0
  # scale_fill_brewer(palette = "Dark2")+
  scale_fill_manual(values = c("Betsileo" = "#1B9E77", "Merina" = "#7570B3", "Betsimisaraka" = "#D95F02
  guides()
```




```
# ggsave("/home/EJC199/Dropbox/malagasy_assortative_mating/map.jpg", width = 7, height = 5, dpi = 500)
```

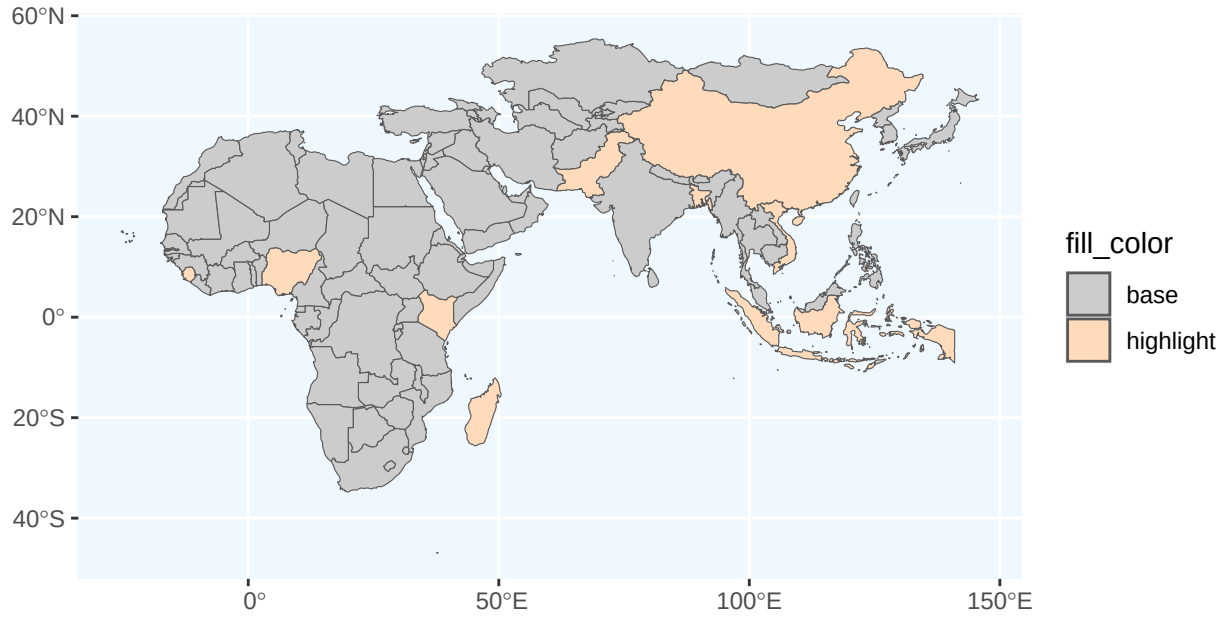
Data currently merged in: Malagasy, 1000 Genomes, Borneo, Sumba

```
# Get all African countries
africa_eurasia <- ne_countries(continent = c("Africa", "Asia"), scale = 50, returnclass = "sf") # %>% s

# Specify countries to highlight and assign colors
```

```r
highlighted_countries <- africa_eurasia %>%
  mutate(fill_color = case_when(
    name %in% c("Madagascar", "Kenya", "Nigeria", "Sierra Leone", "The Gambia", "China", "Vietnam", "Pal
    TRUE ~ "base" # Default color for other countries
  ))


# Plot current data
ggplot() +
  geom_sf(data = highlighted_countries, aes(fill = fill_color)) +
  scale_fill_manual(values = c("highlight" = "#FFDBBB", "base" = "gray80")) +
  # coord_sf(crs = st_crs(29702), xlim = c(-1500000,4800000), ylim = c(-1500000,4500000)) +
  theme(panel.background = element_rect(fill = "#F0F8FF", color = NA))
```



Recording R environment

Part of ensuring our workflow is reproducible is to keep track of the R environment. We will automatically save our R environment into a file called `r_environment.json`.